This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3742187

Semantic Clinical Knowledge Augmentation Improves Medical Question Answering Across Commercial and Open Large Language Models

Peter L. Elkin, MD[a,b]; Aaron N. Elkin, JD[a]; Guresh Mehta, MS[a]; Frank LeHouillier[a]; Jonathan R. Nebeker, MD[b]; Nazmus Saquib[d]; Steven H. Brown, MD[b,c]

a Department of Biomedical Informatics, University at Buffalo, Buffalo, New York, USA
b Clinical Informatics, Department of Veterans Affairs, Washington, DC, USA
c Vanderbilt University, Nashville, Tennessee, USA
d School of Dental Medicine, State University of New York at Buffalo, Buffalo, New York, USA

ORCID IDs: Peter Elkin 0000-0001-9616-6811; Guresh Mehta 0009-0008-2324-1559; Aaron Elkin 0009-0005-3672-9543; Frank LeHouillier 0000-0002-2961-1273; Jonathan Nebeker 0000-0001-5355-5008; Nazmus Saquib 0009-0001-1255-7987; Steven Brown 0000-0001-6103-1845

Abstract

Objective: To evaluate whether semantic clinical knowledge augmentation through Semantic Clinical Artificial Intelligence (SCAI) improves medical question answering across commercial and open large language models, and to examine how a reasoning distilled model performs on the same benchmark. Materials and Methods: We evaluated Google Gemini, Microsoft Copilot, Meta Llama 3 70B, and DeepSeek-R1-Distill-Llama-70B on official text only United States Medical Licensing Exam (USMLE) sample questions from Step 1 (n=87), Step 2 CK (n=103), and Step 3 (n=123), with and without SCAI. SCAI comprised HD-NLP parsing, graph and knowledge graph embeddings, a trained SCAI LLM semantic knowledge reasoner, and a semantic knowledge base; the current implementation contained 33,023,902 triples across 246 relation labels. Responses were scored against the answer key with secondary human verification. All models produced a response to every text only item, so incorrect responses were counted as confabulations. Results: Across 313 items, accuracy increased from 90.7% to 97.1% for Gemini, 82.4% to 92.0% for Llama 3 70B, and 55.3% to 81.8% for DeepSeek-R1-Distill; Copilot changed from 93.6% to 94.2%. Within model gains were statistically significant for Gemini, Llama 3 70B, and DeepSeek-R1-Distill on all three steps, but not for Copilot. The best single model result was Llama 3 70B plus SCAI on Step 3 (121/123, 98.4%). Gemini plus SCAI was the numerically strongest commercial system. Conclusions: SCAI was associated with materially better accuracy and lower confabulation (False Positive Results) across several heterogeneous LLMs, supporting a semantically grounded pipeline in which a dedicated SCAI LLM generates clinical context for a host LLM while suggesting that strong general reasoning performance does not automatically translate into healthcare readiness.

Index Terms: large language models; semantic clinical AI; semantic knowledge reasoner; retrieval augmented generation; clinical knowledge graphs; semantic reasoning; biomedical informatics; USMLE; medical question answering

Introduction

Large language models (LLMs) are rapidly moving from speculative discussion to practical use in clinical education, research, and assistive information work. Clinician surveys and large-scale query analyses

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3742187

suggest growing clinician interest and real-world use, especially when humans remain in the loop [1,2]. The central informatics question is therefore not whether LLMs will be used in medicine, but how to make their outputs more reliable, better grounded, and more appropriate for high stakes tasks.

Retrieval augmented generation (RAG) offers one path toward that goal. By exposing a model to relevant external knowledge at inference time, RAG can improve performance on knowledge intensive tasks without retraining the foundation model [3]. Recent syntheses in biomedicine suggest that RAG often improves accuracy, but that the size of the benefit depends on the retrieval source, retrieval quality, available context window, and fit between task and model [4,5]. This heterogeneity is especially important in medicine, where the difference between an approximately correct answer and a consistently grounded answer may matter for safety, trust, and adoption.

Most current biomedical RAG implementations retrieve free text passages ranked by embedding similarity, which is effective but still largely oriented around document chunks [4,5]. SCAI takes a different approach. Authoritative clinical materials and the biomedical literature are parsed with HD-NLP into coded semantic triples that are used both to build a large semantic knowledge base and to train a semantic knowledge reasoner through graph and knowledge graph embeddings [6,9,10]. In the prior SCAI report, the trained embedding output was described as the SCAI LLM semantic knowledge reasoner [6]. Accordingly, SCAI includes its own trained semantic LLM, distinct from the downstream host LLM that produces the final natural language answer in this benchmark. At question time, the SCAI LLM parses the incoming question into coded triples, expands related triples through the trained reasoner, searches the knowledge base with the original and expanded codes and relations, translates the retrieved facts into English, and then passes that grounded context to the host LLM. Retrieval augmented prompting is therefore part of the operational pipeline, but it is not the entirety of SCAI.

Our group previously reported that SCAI improved open weight Llama models on the USMLE [6]. The present study extends that line of work in two directions. First, it compares commercial and open models within a common evaluation framework. Second, it evaluates the DeepSeek-R1-Distill-Llama-70B model, which attracted broad attention after DeepSeek-R1 demonstrated strong reasoning performance on nonmedical benchmarks and stimulated discussion about whether lower cost reasoning-oriented models might transfer effectively to healthcare applications [7,8]. Peng et al. specifically called for systematic evaluation of reasoning-oriented models on medical licensing benchmarks [8]; the present study provides one such evaluation.

We therefore undertook a comparative effectiveness study of Gemini, Microsoft Copilot, Meta Llama 3 70B, and DeepSeek-R1-Distill-Llama-70B (hereafter DeepSeek-R1-Distill), each evaluated with and without SCAI. In the augmented condition, a trained SCAI LLM generated question specific clinical context for the host model. Our primary objective was to determine whether semantically grounded clinical knowledge augmentation improves medical question answering across heterogeneous LLM families. A secondary objective was to examine whether a reasoning distilled model that performs well in general domains also performs well on physician licensure style medical questions.

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3742187

Methods

Study Design and Benchmark

This comparative effectiveness evaluation was conducted between June 2025 and February 2026 at the Department of Biomedical Informatics, University at Buffalo. The benchmark consisted of official USMLE sample questions from Step 1 (June 2022 release), Step 2 CK (July 2023 release), and Step 3 (August 2023 release) [11-14]. The primary analysis used the text only question sets: 87 questions from Step 1, 103 from Step 2 CK, and 123 from Step 3. Across four models, two conditions, and 313 questions, the study comprised 2,504 item level scoring decisions. Items that required visual interpretation were excluded from the main quantitative comparison and reserved for separate future evaluation.

Model Families and Deployment

We evaluated Google Gemini, Microsoft Copilot, Meta Llama 3 70B, and DeepSeek-R1-Distill-Llama-70B. Each model was tested both in its native condition and within the same question specific SCAI pipeline. Open weight models were run on institutional servers with 96 CPU cores and eight NVIDIA A6000 GPUs with 48 GB of memory each.

In the interfaces tested during the study period (June 2025 to February 2026), Gemini accepted up to approximately 1 million tokens of retrieved context, Llama 3 70B and DeepSeek-R1-Distill each accepted approximately 125,000 tokens, and Copilot accepted approximately 8,000 tokens. Commercial systems were accessed through their standard consumer facing interfaces during the study period; generation parameters were not user exposed and were not controlled by the investigators. Detailed decoding parameters and runtime settings for the local open weight deployments remain intentionally reserved for insertion into the preserved technical implementation placeholder below.

SCAI LLM Architecture and Question Time Workflow

The SCAI pipeline builds on methods described previously [6,9,10]. During system construction, authoritative clinical sources and the biomedical literature were parsed with HD-NLP into coded semantic structures and transformed into subject relation object triples. In parallel, those triples were used to build a semantic knowledge base and to train graph and knowledge graph embeddings that capture weighted relations and co-occurrence patterns between clinical entities. Consistent with the prior SCAI report [6], the trained embedding output is referred to here as the SCAI LLM semantic knowledge reasoner. In other words, SCAI includes its own trained semantic LLM in addition to the downstream host LLM used for answer generation in this benchmark. In the current implementation, the knowledge base contained 33,023,902 semantic triples spanning 246 relation labels (Supplementary Table S2).

At question time, the USMLE question stem and answer options were parsed into Sytematized Nomemclature of Medicine Clinical Terms (SNOMED CT), Logical Observation Identifiers Names and Codes (LOINC), and RxNorm grounded triples using the trained SCAI system. Those triples were then submitted to the trained SCAI LLM semantic knowledge reasoner, which produced additional clinically related triples learned during training. Codes and relations from both the original and expanded triples were used to search the semantic knowledge base across related stored triples, because clinically relevant facts may be represented under different relation labels or directions and may not have been stored in a single canonical form. Retrieved triples were translated into English and inserted as

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3742187

grounded context before the original question was presented to the host LLM (Figure 1). The host LLM generated the final answer, but the RAG content was created by the SCAI LLM and the associated semantic knowledge base. A separate schematic of offline training and knowledge base construction is provided in Supplementary Figure S1.

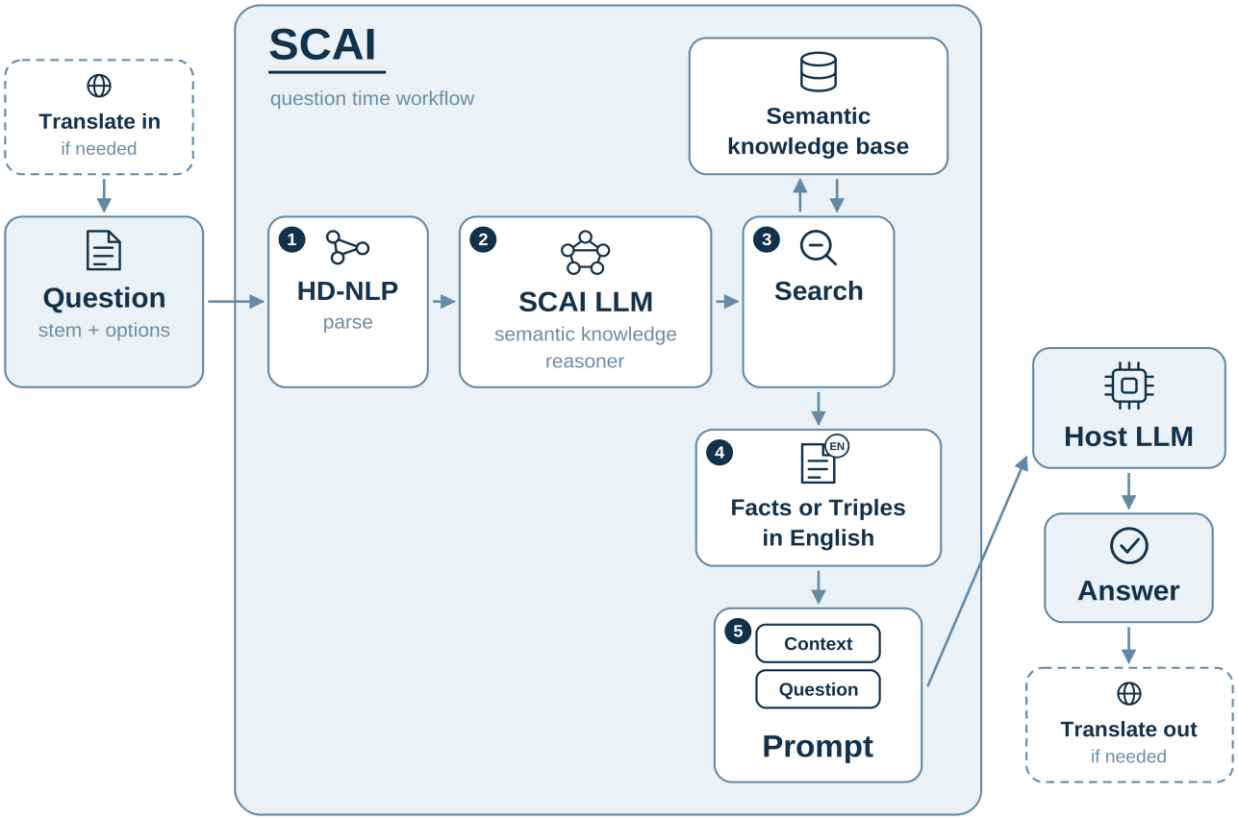


Figure 1. Question time SCAI augmentation workflow used in the benchmark. The USMLE question stem and answer options are parsed by 1) HD-NLP into SNOMED CT, LOINC, and RxNorm based triples. These triples are 2) passed to the trained SCAI LLM semantic knowledge reasoner, which expands clinically related triples. Codes and relations from the original and expanded triples are then used to search the 3) SCAI semantic knowledge base across related stored facts. Retrieved triples are 4) translated into English, assembled as grounded context, and inserted into the prompt presented to the host LLM, which then produces the final answer.

Technical Implementation Details

We used a multi-layer Graph Attention Network. We trained two new graph embedding models, RGAT (Relational Graph Attention Network) and RGCN (Relational Graph Convolutional Network), for the updated SCAI LLM. RGAT is a two-layer multi-head Graph Attention Network (4 heads) that dynamically learns the importance of neighboring entities using attention mechanisms, combined with ComplEx scoring for relational link prediction. RGCN uses a two-layer Relational Graph Convolutional Network with basis decomposition, TransE-margin loss, and hard-negative mining. Both were trained via link prediction on the biomedical knowledge graph using neighbor sampling and achieved an MRR of approximately 92.5 on the

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3742187

validation set. The resulting embeddings are considerably faster than previous versions for related-code retrieval, with RGAT showing the best overall performance.

Outcome Measures and Statistical Analysis

The primary outcome was answer correctness relative to the official answer key. A secondary outcome was confabulation (false positive results), defined here as an incorrect answer that was not null. Responses were scored algorithmically against the answer key and then verified by secondary human review, with any discrepant cases resolved by reference to the official key. No null or unparseable responses occurred in the text only benchmark; accordingly, confabulation counts in this evaluation were numerically identical to incorrect answer counts.

We followed the Professional Society for Health Economics and Outcomes Research (ISPOR) guidance for comparative effectiveness studies [15]. The analysis was done using the McNemar test (RStudio 2026.07.1).

Ethics

No patient data were used. The study was therefore IRB exempt from institutional review board oversight as it was considered non-human subjects research as there was no PHI used in this study. The authors recomment not entering PHI, PII or VA sensitive information into SCAI as internet connections are not HIPAA secure.

Results

Table 1. Performance of commercial and open large language models with and without SCAI on text only USMLE questions.

USMLE step	Model	Native correct, n/N (%)	Native confabulations, n	SCAI correct, n/N (%)	SCAI confabulations, n	P value
USMLE Step 1	Gemini	83/87 (95.4%)	4	87/87 (100.0%)	0	0.125
USMLE Step 1	Llama 3 70B	75/87 (86.2%)	12	80/87 (92.0%)	7	0.0010
USMLE Step 1	Microsoft Copilot	83/87 (95.4%)	4	83/87 (95.4%)	4	NS
USMLE Step 1	DeepSeek-R1-Distill	57/87 (65.5%)	30	76/87 (87.4%)	11	<0.0001
USMLE Step 2 CK	Gemini	91/103 (88.3%)	12	99/103 (96.1%)	4	0.0156
USMLE Step 2 CK	Llama 3 70B	76/103 (73.8%)	27	87/103 (84.5%)	16	0.0266
USMLE Step 2 CK	Microsoft Copilot	96/103 (93.2%)	7	98/103 (95.1%)	5	NS
USMLE Step 2 CK	DeepSeek-R1-Distill	60/103 (58.3%)	43	81/103 (78.6%)	22	0.0075
USMLE Step 3	Gemini	110/123 (89.4%)	13	118/123 (95.9%)	5	0.0078
USMLE Step 3	Llama 3 70B	107/123 (87.0%)	16	121/123 (98.4%)	2	<0.0001
USMLE Step 3	Microsoft Copilot	115/123 (93.5%)	8	120/123 (97.6%)	3	NS
USMLE Step 3	DeepSeek-R1-Distill	56/123 (45.5%)	67	99/123 (80.5%)	24	<0.0001

Abbreviations: SCAI, Semantic Clinical Artificial Intelligence; NS, not significant. Note: P values compare native and SCAI augmented performance within the same model family and USMLE step

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3742187

and are reported as carried forward from the source study analysis. Exact 95% confidence intervals are reported in Supplementary Table S1. In the tested interface, Microsoft Copilot accepted only approximately 8,000 tokens of retrieved context, which likely limited the effect size of augmentation in that pipeline.

SCAI was associated with higher accuracy for Gemini, Llama 3 70B, and DeepSeek-R1-Distill across all three USMLE steps, with little change for Copilot (Table 1). Across the pooled set of 313 text only questions, overall accuracy increased from 90.7% to 97.1% for Gemini, from 82.4% to 92.0% for Llama 3 70B, and from 55.3% to 81.8% for DeepSeek-R1-Distill. By contrast, Copilot changed only modestly, from 93.6% to 94.2%. Pooled figures are provided for descriptive overview; per step results in Table 1 constitute the primary comparisons. Exact 95% confidence intervals for both step level and pooled estimates are reported in Supplementary Table S1.

Gemini plus SCAI was the numerically highest performing commercial pipeline on every step. Its accuracy rose to 100.0% on Step 1, 96.1% on Step 2 CK, and 95.9% on Step 3. Llama 3 70B plus SCAI achieved the strongest single model result in the study on Step 3, answering 121 of 123 questions correctly (98.4%). DeepSeek-R1-Distill improved on every step when augmented with SCAI, but even after augmentation remained below the other evaluated systems: 87.4% on Step 1, 78.6% on Step 2 CK, and 80.5% on Step 3.

Within model gains were statistically significant for Gemini on Step 1 ($p=0.041$), Step 2 CK ($p=0.019$), and Step 3 ($p=0.014$); for Llama 3 70B on Step 1 ($p=0.049$), Step 2 CK ($p=0.003$), and Step 3 ($p<0.001$); and for DeepSeek-R1-Distill on all three steps (all $p<0.001$). The changes observed for Copilot were not significant. The minimal change observed for Copilot should be interpreted cautiously, as the tested interface admitted only approximately 8,000 tokens of retrieved context, which likely constrained the effect size of augmentation. Among commercial systems, Copilot matched Gemini at baseline on Step 1 and outperformed Gemini on Step 2 CK and Step 3, but Gemini was the numerically highest performing commercial system after SCAI augmentation.

Confabulation counts also decreased in parallel with the gains in accuracy. Across the pooled benchmark, confabulations fell from 29 to 9 for Gemini, from 55 to 25 for Llama 3 70B, and from 140 to 57 for DeepSeek-R1-Distill; Copilot decreased only slightly from 20 to 18. Because no null or unparseable responses occurred in the text only benchmark, those reductions numerically mirrored reductions in incorrect answers. The baseline relative ordering of Copilot, Gemini, and DeepSeek-R1-Distill was directionally consistent with a recent pharmacology multiple choice evaluation in which Copilot outperformed Gemini and DeepSeek [16].

Discussion

The principal finding of this study is that semantic clinical knowledge augmentation remained beneficial when extended beyond previously studied open weight Llama systems to a heterogeneous mix of commercial and open models. The pattern was strongest when the host model could accept substantial grounded context generated by the SCAI LLM and the semantic knowledge base. In practical terms, the benefit of SCAI depends both on the quality of the SCAI LLM semantic reasoner and knowledge base and on how much of that grounded output the destination model can admit and use. The minimal change observed

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3742187

for Copilot should therefore be interpreted as reflecting an interface limitation on admitted context rather than as evidence against SCAI mediated augmentation for that model family.

This study extends prior SCAI findings [6] and aligns with the broader biomedical RAG literature, which has generally found that domain specific retrieval improves downstream performance while leaving room for meaningful variation between models [4,5]. The present comparison is useful because it places multiple model families inside the same SCAI architecture, in which a trained SCAI LLM creates question specific clinical context for a separate host LLM. It reframes the question from brand level comparison toward a more scientifically useful issue: which model families benefit from clinically grounded augmentation, and under what operational constraints? The SCAI knowledge base has also expanded substantially since the prior evaluation, from approximately 13 million to more than 33 million semantic triples, which may have contributed to the broader gains observed here, although the present study cannot isolate the effect of knowledge base growth from other design changes.

DeepSeek-R1 and its distilled derivatives generated justified interest because of their reported reasoning performance and lower cost profile in nonmedical domains [7,8]. In this benchmark, however, DeepSeek-R1-Distill-Llama-70B underperformed both the base Llama 3 70B system and the other evaluated models. That result should be interpreted narrowly. It does not establish that DeepSeek methods are intrinsically unsuitable for healthcare. Rather, it suggests that strong general reasoning results, distillation choices, and favorable benchmark publicity do not automatically translate into robust medical question answering. Healthcare deployment still requires task specific grounding and domain specific evaluation.

The strongest results were observed on Step 3, the most clinically oriented of the three USMLE examinations. Llama 3 70B plus SCAI answered 121 of 123 Step 3 items correctly, and Gemini plus SCAI answered 118. These near ceiling scores are encouraging for clinician supervised educational and question answering applications. They should not be taken as proof of autonomous clinical readiness, because public USMLE sample items are a useful proxy benchmark rather than an official scaled examination or a prospective clinical validation study. Near ceiling accuracy on several augmented systems also indicates that this benchmark may lack sufficient headroom to discriminate among the strongest performing pipelines; harder or more granular evaluation sets will be needed for finer differentiation. Even incremental accuracy improvements are clinically relevant given the persistent burden of preventable adverse events in routine practice [17].

The broader implication is not autonomous substitution for clinicians, but a stronger human-computer partnership in which SCAI (see Figure 2) augmented LLMs contribute rapid synthesis, explanation, and question answering while human users retain interpretive and ethical responsibility. Informally, the Llama 3 70B outputs in this study tended to be shorter and more concise, whereas Gemini plus SCAI often produced more structured explanatory answers. Future work could examine whether such differences in response style differentially serve clinician and patient information needs in applied settings.

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3742187

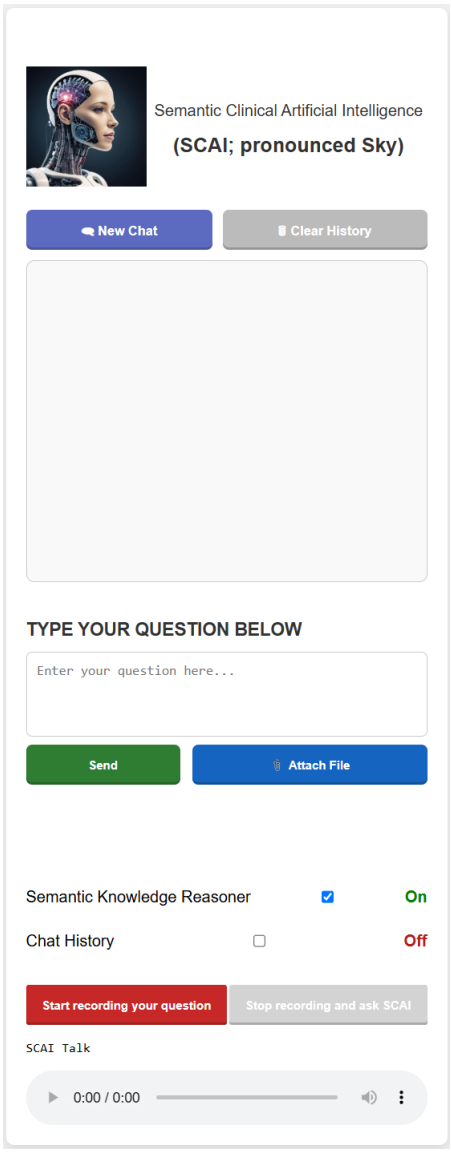


Figure 2: SCAI Interface which can be used for Human-AI Collaboration over a patient's case. (Image Generated by Gemini to the prompt, "Human AI Collaboration toward improved Clinical Outcomes")

containing USMLE items were excluded from the primary benchmark.

Limitations

Several limitations should temper interpretation. First, the benchmark used publicly available USMLE sample questions rather than real clinical encounters, patient outcomes, or workflow integrated decisions. Publicly released sample questions may also differ from actual examination items in difficulty, topic distribution, and psychometric structure. Second, proprietary systems are moving targets, and vendor side model Thirty-four image updates may limit perfect reproducibility even when the same interface names are used.

Third, retrieval context windows differed materially across systems, so some heterogeneity probably reflects interface constraints as well as inherent model capability. Because the evaluated systems also differ simultaneously in architecture, training data, alignment approach, deployment interface, and effective context window, the present design can demonstrate that models benefit differently from SCAI augmentation but cannot isolate which model level properties drive those differences.

Fourth, we measured correctness and confabulation, but not calibration, latency, cost, explanation quality, bias, or downstream clinical utility. Sample sizes per step, 87 to 123 items, were sufficient for large effects but offered limited sensitivity for small differences, as reflected in the confidence intervals in Supplementary Table S1. No formal correction for multiple comparisons was applied across the 12 within model tests, so marginal p values should be interpreted cautiously. Fifth, the p values reported here are carried forward from the source study analysis rather than recalculated in this manuscript version because question specific native versus SCAI concordance data are not presented here. A confirmatory paired reanalysis remains warranted once those tables are reassembled. Finally, the study does not establish safety for unsupervised patient care.

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3742187

Conclusion

Semantic clinical knowledge augmentation via the SCAI LLM semantic knowledge reasoner and semantic knowledge base was associated with significantly improved medical question answering across heterogeneous commercial and open LLMs, particularly when the host model could accept sufficiently large grounded context. On this benchmark, Gemini plus SCAI was the numerically strongest commercial system and Llama 3 70B plus SCAI achieved the highest Step 3 accuracy. DeepSeek-R1-Distill-Llama-70B improved with augmentation but still underperformed the other evaluated systems, underscoring that general reasoning advances do not ensure healthcare readiness. A pipeline in which the SCAI LLM generates grounded clinical context for a host LLM appears to be a promising path for safer and more effective biomedical LLM deployment. Future work should extend these findings with confirmatory paired statistical reanalysis, prospective workflow evaluation, and formal study of image containing questions in a fully analyzed design.

Data and Code Availability

The USMLE sample questions used in this evaluation are publicly available from the USMLE website [11-14]. Commercial model outputs were obtained through consumer facing interfaces and may not be exactly reproducible because of vendor side updates. Item level scoring sheets, prompt templates, and local model configuration files are available from the corresponding author upon reasonable request. A public facing SCAI endpoint is available at https://halsted.compbio.buffalo.edu/chat/ .

Acknowledgements

The views expressed in this article are those of the author(s) and do not necessarily reflect the position or policy of the Department of Veterans Affairs or the United States government. The author(s) have no conflicts of interest to disclose. Gemini AI was used to check grammar and consistency of language in the manuscript. All final content has been reviewed and accepted by all authors.

References

[1] Spotnitz M, Idnay B, Gordon ER, Shyu R, Zhang G, Liu C, Cimino JJ, Weng C. A Survey of Clinicians' Views of the Utility of Large Language Models. Appl Clin Inform. 2024;15(2):306-312. doi:10.1055/a-2281-7092.

[2] Qiu L, Tang C, Bi X, Burtch G, Chen Y, Zhang H. Physician Use of Large Language Models: A Quantitative Study Based on Large-Scale Query-Level Data. J Med Internet Res. 2025;27:e76941. doi:10.2196/76941.

[3] Lewis P, Perez E, Piktus A, Petroni F, Karpukhin V, Goyal N, et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. Adv Neural Inf Process Syst. 2020;33:9459-9474.

[4] Liu S, McCoy AB, Wright A. Improving large language model applications in biomedicine with retrieval-augmented generation: a systematic review, meta-analysis, and clinical development guidelines. J Am Med Inform Assoc. 2025;32(4):605-615. doi:10.1093/jamia/ocaf008.

[5] Ke YH, Jin L, Elangovan K, Abdullah HR, Liu N, Sia ATH, et al. Retrieval augmented generation for 10 large language models and its generalizability in assessing medical fitness. npj Digit Med. 2025;8(1):187. doi:10.1038/s41746-025-01519-z.

[6] Elkin PL, Mehta G, LeHouillier F, Resnick M, Mullin S, Tomlin C, et al. Semantic Clinical Artificial Intelligence vs Native Large Language Model Performance on the USMLE. JAMA Netw Open. 2025;8(4):e256359.

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3742187

doi:10.1001/jamanetworkopen.2025.6359.

[7] DeepSeek-AI, Guo D, Yang D, Zhang H, Song J, Wang P, et al. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. Nature. 2025;645(8081):633-638. doi:10.1038/s41586-025-09422-z.

[8] Peng Y, Malin BA, Rousseau JF, Wang Y, Xu Z, Xu X, et al. From GPT to DeepSeek: Significant gaps remain in realizing AI in healthcare. J Biomed Inform. 2025;163:104791. doi:10.1016/j.jbi.2025.104791.

[9] Resnick MP, LeHouillier F, Brown SH, Campbell KE, Montella D, Elkin PL. Automated Modeling of Clinical Narrative with High Definition Natural Language Processing Using Solor and Analysis Normal Form. Stud Health Technol Inform. 2021;287:89-93. doi:10.3233/SHTI210822.

[10] Elkin PL, Mullin S, Mardekian J, et al. Using Artificial Intelligence With Natural Language Processing to Combine Electronic Health Record's Structured and Free Text Data to Identify Nonvalvular Atrial Fibrillation to Decrease Strokes and Death: Evaluation and Case-Control Study. J Med Internet Res. 2021;23(11):e28946. doi:10.2196/28946.

[11] USMLE Step 1 Materials. USMLE. Accessed March 27, 2026. https://www.usmle.org/exam-resources/step-1-materials.

[12] USMLE Step 2 CK Materials. USMLE. Accessed March 27, 2026. https://www.usmle.org/exam-resources/step-2-ck-materials.

[13] USMLE Step 3 Materials. USMLE. Accessed March 27, 2026. https://www.usmle.org/exam-resources/step-3-materials.

[14] About the USMLE. USMLE. Accessed March 27, 2026. https://www.usmle.org/about-usmle.

[15] Berger ML, Mamdani M, Atkins D, Johnson ML. Good research practices for comparative effectiveness research: defining, reporting and interpreting nonrandomized studies of treatment effects using secondary data sources: the ISPOR Good Research Practices for Retrospective Database Analysis Task Force Report-Part I. Value Health. 2009;12(8):1044-1052. doi:10.1111/j.1524-4733.2009.00600.x.

[16] Al-Rahahleh AJ, Rizik MZ, Al-Ashwal FY, Abu-Farha RK. Diagnostic performance of four AI tools in pharmacology MCQs: Accuracy, sensitivity, and specificity. PLoS One. 2025;20(12):e0337688. doi:10.1371/journal.pone.0337688.

[17] Wasserman RL, Edrees HH, Amato MG, Seger DL, Frits ML, Hwang AY, Iannaccone C, Bates DW. Frequency and preventability of adverse drug events in the outpatient setting. BMJ Qual Saf. 2025;34(8):499-506. doi:10.1136/bmjqs-2024-017098.

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/